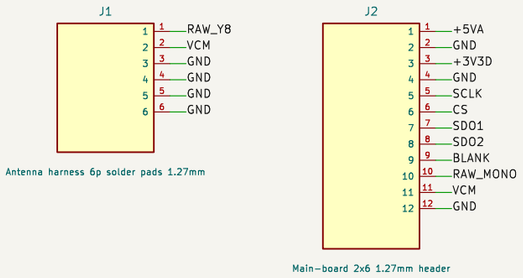
PF233-AB1B: AB1B receive analog board: 2 channels (Y8, MONO), blanking, gain, 16-bit ADC

How to read: parts sit in functional blocks (dashed frames). Every pin ends in a short wire with its NET NAME; equal names are connected (also between blocks). A cross = intentionally unconnected pin.
Diode symbols point anode to cathode. Pin names are those of the library; items marked VERIFY in docs/ASSUMPTIONS.md are unverified. MODELED design: not built, not bench-tested.

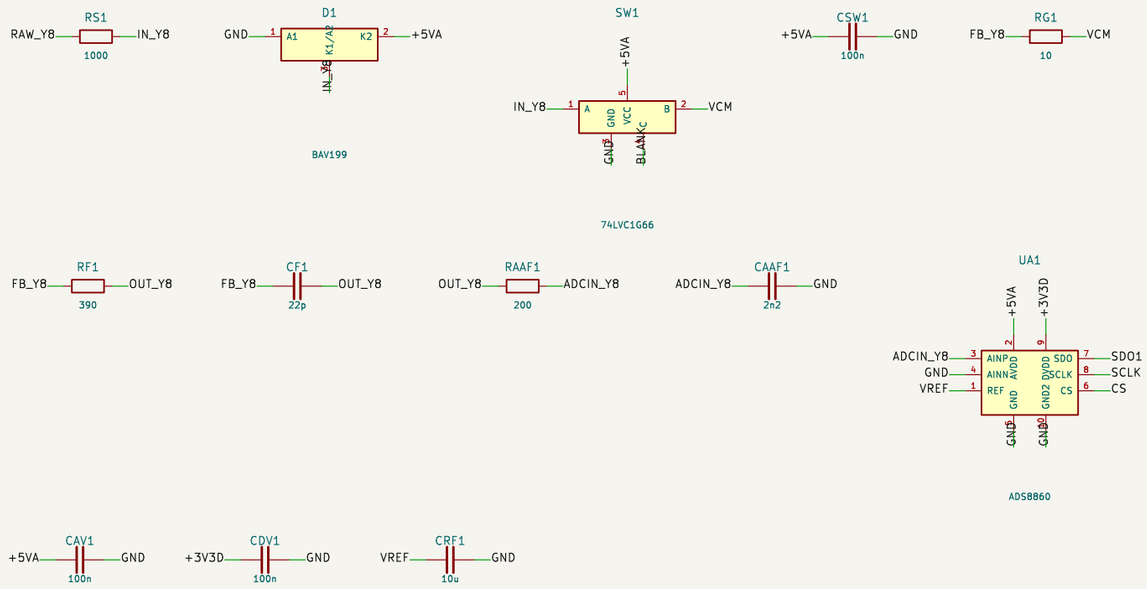
1 Connectors

J1: antenna harness (raw coil signals RAW_*, and VCM returns).
J2: main-board 2x6 header: +5VA, +3V3D, SPI (SCLK, CS, SDO*), BLANK, VCM.



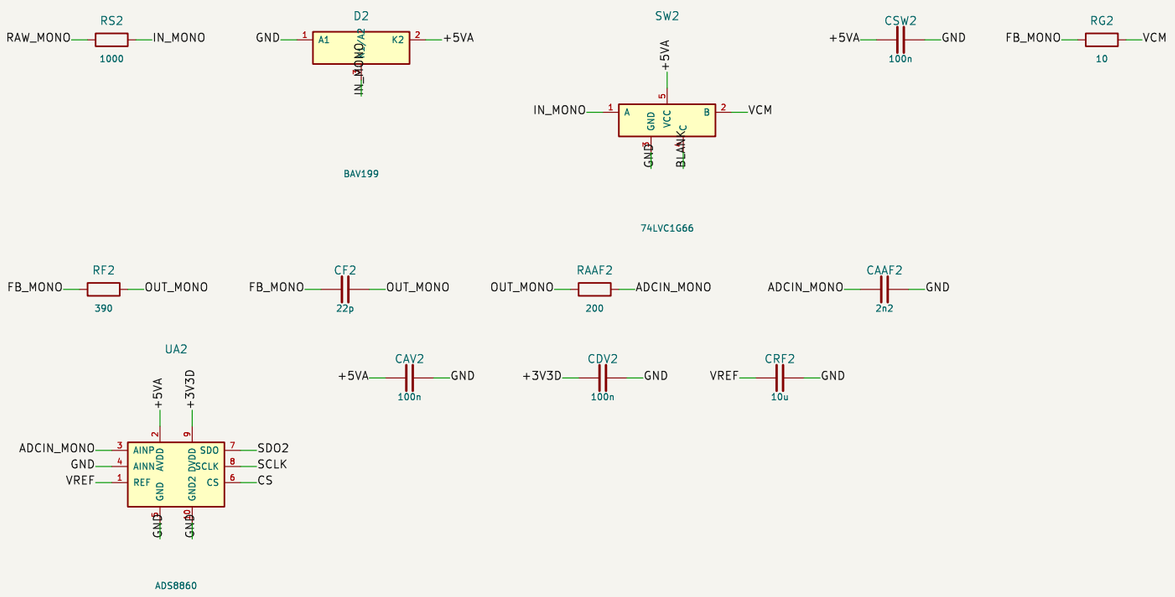
2 Channel Y8

RAW_Y8 -> RS1 (series) -> D1 (BAV199 clamp to +5VA / GND) -> SW1 (74LVC1G66; BLANK opens the path) -> IN_Y8 -> op-amp U1 half A.
Non-inverting gain stage: RG1 to VCM, RF1 || CF1 feedback; OUT_Y8 -> RAAF1 / CAAF1 anti-alias -> ADCIN_Y8 -> UA1 (ADS8860, 16-bit SAR; REF = VREF with CRF1).
SPI: SCLK / CS from the main board, SDO1 back.



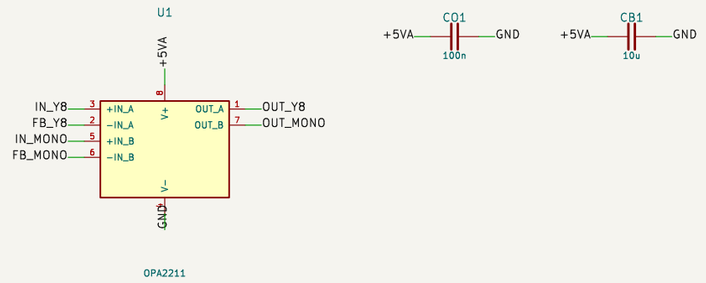
3 Channel MONO

RAW_MONO -> RS2 (series) -> D2 (BAV199 clamp to +5VA / GND) -> SW2 (74LVC1G66; BLANK opens the path) -> IN_MONO -> op-amp U1 half B.
Non-inverting gain stage: RG2 to VCM, RF2 || CF2 feedback; OUT_MONO -> RAAF2 / CAAF2 anti-alias -> ADCIN_MONO -> UA2 (ADS8860, 16-bit SAR; REF = VREF with CRF2).
SPI: SCLK / CS from the main board, SDO2 back.



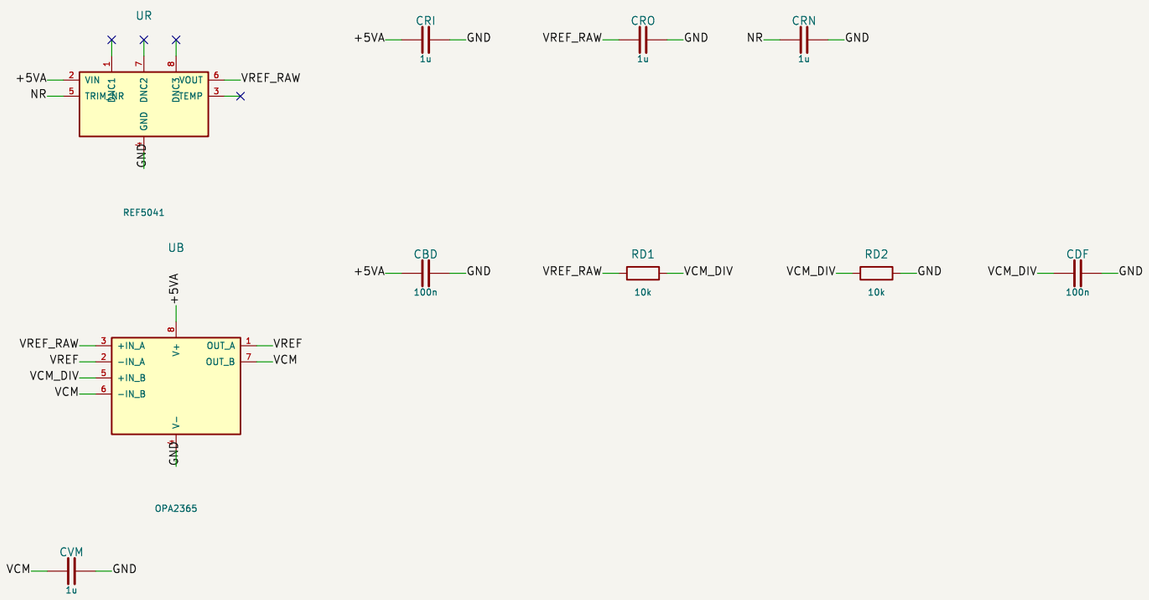
4 Gain-stage op-amps

OPA2211 dual op-amps (each half = one channel) from +5VA, decoupled CO*; CB1 bulk.



5 Reference and common-mode

UR (REF5041) -> VREF_RAW (CRI / CRO / CRN) -> UB (OPA2365) buffers VREF for the ADCs; RD1 / RD2 / CDF divide VREF_RAW to VCM_DIV -> UB other half -> VCM (CVM).



Generated: src/pcb/sch_func.py; checked against the netlist.
MODELED design: not built, not bench-tested.
AB1B receive analog board: 2 channels (Y8, MONO), blanking, gain, 16-bit ADC
ARX / PHI-FOLD 233

Sheet: /
File: PF233-AB1B_functional.kicad_sch

Title: PF233-AB1B functional schematic

Size: User	Date: 2026-10-26	**Rev: r3**
KiCad E.D.A. 9.0.9		Id: 1/1